

Explainable Transformer-Based Multi-Modal Anomaly Detection for Secure Vehicle-to-Grid Charging Stations

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Abstract—Vehicle-to-Grid (V2G) charging stations are becoming critical cyber-physical components in modern electric mobility infrastructure. Their bidirectional connectivity with vehicles, backend systems, and the power grid exposes them to a wide range of cyberattacks that manifest differently in network traffic and host-side behavior, making reliable detection inherently multi-modal. Detecting these attacks reliably requires a system that can reason over both data streams jointly while resisting the inflated performance estimates that arise when training and test windows share the same continuous capture sessions. We propose an explainable transformer-based anomaly detection system for electric vehicle supply equipment (EVSE) that addresses these challenges together. Our system operates on raw event tokens rather than aggregated bins, grouping each 30-second window into up to 32 host tokens and 64 network tokens. These tokens are projected into a shared latent space, enriched with modality embeddings and timestamp positional encoding, and processed jointly by a four-layer transformer encoder for five-class classification spanning Benign, Reconnaissance, DoS, Backdoor, and Cryptojacking behaviors. To obtain a defensible performance estimate, we adopt a strict session-level holdout that assigns complete capture sessions disjointly to training, validation, and test sets. Under this protocol, our model achieves 90.30% accuracy and 0.6952 macro F1 on 701 test windows, with strong detection of Reconnaissance, Backdoor, and Cryptojacking. The evaluation also reveals a benign cross-device failure caused by distribution shift between EVSE environments — a finding that looser splits would have concealed. Explanation is provided through SHAP feature attribution over 18,944 attribution dimensions and attention map analysis, making the detector auditable for operational use.

Index Terms—Vehicle-to-Grid, EVSE cybersecurity, anomaly detection, transformer, multi-modal fusion, explainable AI, SHAP, CICEVSE2024

I. INTRODUCTION

Electric vehicles are rapidly becoming part of everyday transportation, and the charging infrastructure supporting them is expanding at a comparable pace [1]. In Vehicle-to-Grid (V2G) systems, a charging station does more than deliver energy to a vehicle: it participates in bidirectional energy exchange with the grid, communicates with cloud-based management systems, and runs software on embedded hardware continuously exposed to the network [2]. This makes electric vehicle supply equipment (EVSE) a high-value target whose compromise can disrupt service continuity, degrade grid sta-

bility, and undermine operational trust in the broader V2G ecosystem.

Detecting attacks against an EVSE is difficult for several structural reasons. First, the data is multi-modal: some attacks are primarily visible in network flows, while others — such as persistent host-level compromise or unauthorized cryptocurrency mining — leave clearer traces in hardware performance counters and kernel events. Second, the data is temporally structured, and detecting attack patterns requires reasoning over event sequences rather than individual observations. Third, and perhaps most consequentially, the way train and test splits are constructed can silently inflate reported performance: when windows from the same continuous capture session appear on both sides of the split, a model can succeed by memorizing session structure rather than learning to detect attacks.

Existing work on EVSE cybersecurity has made useful progress, but it consistently leaves at least one of these challenges unaddressed. Most studies examine a single modality, reduce the attack taxonomy to fewer classes, or adopt evaluation protocols that do not prevent session-level overlap between training and testing. As a result, it is difficult to judge how a detector would actually perform once capture conditions change.

We address these challenges through an explainable multi-modal transformer built on raw host and network event tokens, trained and evaluated under a strict session-level holdout. The remainder of the paper reviews the relevant literature (Section II), states the contributions (Section III), defines the system model and problem formulation (Section IV), presents the proposed approach (Section V), discusses results and analysis (Section VI), and concludes (Section VII).

II. LITERATURE REVIEW

A. V2G Intrusion Detection on CICEVSE2024

Detection of cyberattacks in V2G environments remains less studied than broader communication security and protection mechanisms [3], [4]. The CICEVSE2024 dataset [5] is the primary benchmark in this space, providing both network traffic flows and host-side hardware performance counters and kernel events — a combination that makes it possible to study attacks that span both modalities. However, existing work

on this dataset consistently restricts scope in ways that limit what can be concluded about full EVSE behavior. Buedi et al. [5] evaluate the two modalities separately under random cross-validation, leaving joint modeling unexplored. Rahman et al. [6] achieve strong results on a 3-class network-only task but exclude Backdoor and Cryptojacking, the two attack classes that are only observable in host data. Makhmudov et al. [7] further collapse all attacks into a single binary label, which precludes any per-class analysis. None of these studies examine what happens to detection performance when training and test windows are drawn from disjoint capture sessions.

B. Transformers and Explainability

Transformer models have demonstrated strong capability for anomaly detection in temporally structured data, primarily because self-attention can capture dependencies across heterogeneous event sequences without relying on fixed-length aggregation windows [8]–[10]. Explainable AI is increasingly relevant in cybersecurity contexts where model predictions may support operational decisions and incident investigation [11]–[13]. In the EVSE setting, however, neither direction has been pursued in combination with multi-modal data: existing work does not bring temporal sequence modeling, host-network fusion, and feature attribution together in a single pipeline evaluated under realistic session conditions.

C. Open Challenges

What remains missing is not a better algorithm in isolation, but a coherent combination of design choices that the literature has not yet assembled in one place. The EVSE detection problem needs a five-class detector that treats host and network evidence as equally informative; a sequence model that preserves temporal detail rather than over-smoothing it into coarse bins; and an evaluation protocol that forces the model to generalize to sessions it has never seen. Without all three, reported results remain difficult to interpret and unlikely to reflect real deployment behavior. Table I positions the most closely related work against these requirements.

III. CONTRIBUTIONS

This paper makes the following contributions.

- 1) We propose a token-level EVSE anomaly detection pipeline that operates on raw host events and network flows grouped into 30-second windows, preserving the temporal variation that coarse aggregation removes.
- 2) We design a transformer encoder that jointly processes both modalities through modality embeddings, timestamp-based positional encoding, and shared self-attention — without requiring separate per-modality models or late-fusion voting.
- 3) We integrate explainability as a first-class component: SHAP attribution over 18,944 dimensions, exported attention maps, and modality-masked inference are all part of the detection workflow, making individual predictions auditable.

TABLE I
COMPACT COMPARISON WITH PRIOR CICEVSE2024 STUDIES

Work	Modality	Classes	Fusion	XAI	Main limitation
Buedi [5]	Separate host / net	5, 2	✗	✗	Random CV and no joint modeling
Rahman [6]	Network only	3	✗	✓	Drops Backdoor and Cryptojacking
Makhmudov [7]	Network only	2	✗	✗	Collapses all attacks into one class
Ours	Host + net	5	✓	✓	Dataset still limits host-attack generalization

- 4) We adopt a strict session-level holdout evaluation that assigns complete capture sessions disjointly across splits, revealing a benign cross-device failure that looser evaluation protocols would have concealed.

IV. SYSTEM MODEL

A. Environment and Threat Model

We consider a V2G charging environment composed of an electric vehicle, an EVSE charging station, a charging station management system, and the power grid. In the underlying testbed, EV–EVSE communication follows the V2G stack while EVSE–backend communication is carried through networked management channels [5]. The EVSE is therefore exposed to attacks at multiple layers: external traffic manipulation, resource exhaustion, reconnaissance activity, and compromise of the host machine itself.

The detector targets five operating classes:

- 1) **Benign**: normal idle or charging operation.
- 2) **Reconnaissance**: scanning and probing behavior such as port scans, service discovery, and OS fingerprinting.
- 3) **DoS**: flooding and resource-exhaustion traffic.
- 4) **Backdoor**: persistent unauthorized remote access on the EVSE host.
- 5) **Cryptojacking**: unauthorized cryptocurrency mining running on the EVSE host.

B. Data Representation

The study uses CICEVSE2024 [5], which provides two public modalities. The network modality contains flow-level features extracted from EVSE traffic, while the host modality contains hardware performance counters and kernel-event traces collected from the EVSE host. After preprocessing and alignment, the fused dataset contains 17,279 labeled rows and 705 retained features: 204 network features and 501 host features.

Our detector does not operate on flat rows. Instead, we construct a 30-second window from raw events and represent it as a sequence of typed tokens. The 30-second duration is chosen to capture a meaningful episode of attack or benign activity while remaining compact enough for batch training; the token budgets — up to 64 network tokens and 32 host tokens per window — reflect the relative event density of each modality in the dataset. Host tokens represent host-side event observations; network tokens represent network-flow observations. Tokens are padded when fewer events are available, and a padding mask prevents the transformer from attending to empty positions. Complete capture sessions are assigned disjointly to training, validation, and test sets, ensuring that the model must generalize across sessions rather than interpolate within them.

C. Problem Formulation

Let $\mathcal{W} = (\mathbf{H}, \mathbf{N})$ denote one 30-second window, where $\mathbf{H}$ is the ordered set of host tokens and $\mathbf{N}$ is the ordered set of network tokens. The detector learns a function

$$f_{\theta}(\mathcal{W}) = \mathbf{p}, \quad \mathbf{p} \in [0, 1]^5, \quad \sum_{c=1}^5 p_c = 1 \quad (1)$$

that returns a five-dimensional class-probability vector over the labels {Benign, Reconnaissance, DoS, Backdoor, Cryptojacking}. The predicted class is

$$\hat{y} = \arg \max_c p_c. \quad (2)$$

The five class indices correspond to $c \in \{0 = \text{Benign}, 1 = \text{Reconnaissance}, 2 = \text{DoS}, 3 = \text{Backdoor}, 4 = \text{Cryptojacking}\}$, spanning both network-visible attacks (classes 1–2) and host-resident behaviors (classes 3–4) that are invisible to network-only detectors. For each prediction, the system also produces an explanation bundle

$$\mathcal{E} = (\Phi_{\text{SHAP}}, \mathbf{A}_{\text{attn}}, \Delta_{\text{mod}}) \quad (3)$$

where Φ_{SHAP} denotes per-feature SHAP attributions computed over the full 18,944-dimensional token representation, $\mathbf{A}_{\text{attn}}$ denotes attention weight matrices exported from the final encoder layer, and Δ_{mod} captures the shift in predicted probabilities when host or network tokens are zeroed out — providing a coarse, interpretable estimate of each modality’s contribution to the decision.

V. PROPOSED APPROACH

The final system follows four stages: token construction, modality-aware encoding, transformer fusion, and explainable classification. Figure 1 summarizes the pipeline.

A. Token Construction

Earlier preprocessing aggregated events into coarse 5-second bins, which reduced variability and made different windows from the same session look too similar. The final workflow therefore preserves raw-event granularity. For each 30-second window, the pipeline collects the most recent host and network observations up to the configured token budgets.

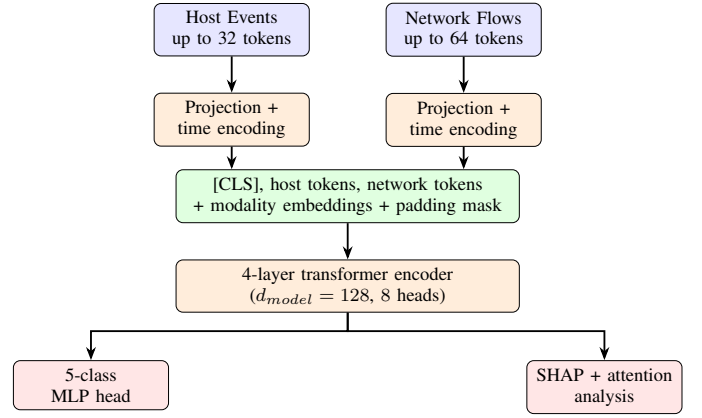


Fig. 1. Final token-fusion architecture used in phase 3.

Each token retains its feature vector together with its timestamp so that the model can reason over event order rather than only event counts.

B. Modality-Aware Encoding

Host and network tokens are first projected into a shared latent space. A special [CLS] token is prepended for sequence-level classification. The input sequence is then enriched with two kinds of structural information:

- 1) **Modality embeddings**, which tell the encoder whether a token originated from the host stream or the network stream.
- 2) **Timestamp positional encoding**, derived from real event times in milliseconds rather than artificial uniform indices.

Padding masks are applied so that empty host or network slots do not attract attention.

C. Transformer Fusion and Classification

The model uses a transformer encoder with four layers, embedding size 128, eight attention heads, feed-forward dimension 256, dropout 0.1, and an MLP classification head. Cross-modal fusion is achieved through self-attention over the joint token sequence rather than through late voting between two separate models. The network is trained with AdamW at learning rate 10^{-4} , batch size 256, and early stopping on validation macro F1.

D. Explainability

Explainability is integrated as part of the workflow rather than as an afterthought. The SHAP module operates over an attribution space of 18,944 dimensions, covering host features across host time steps, network features across network time steps, and the corresponding masking and temporal channels. This makes it possible to attribute a decision to concrete signals such as cache misses, CPU-cycle behavior, context switching, or network-flow statistics. Attention maps are exported from the final encoder layer to show which temporal regions the model emphasizes, while masked-modality analysis

TABLE II
HEADLINE SESSION-LEVEL RESULT OF THE FINAL TOKEN-FUSION MODEL

Metric	Value
Accuracy	90.30%
Macro precision	0.6415
Macro recall	0.7917
Macro F1	0.6952
Macro AUC	0.9175
Test windows	701

TABLE III
PER-CLASS RESULTS UNDER SESSION-LEVEL HOLDOUT

Class	n	Prec.	Rec.	F1
Benign	66	0.000	0.000	0.000
Reconnaissance	55	0.818	0.982	0.893
DoS	43	0.447	0.977	0.613
Backdoor	488	1.000	1.000	1.000
Cryptojacking	49	0.942	1.000	0.970

provides a coarse estimate of how much the decision depends on host versus network evidence.

VI. RESULTS AND ANALYSIS

A. Experimental Setup

The final evaluation uses a session-level holdout. Each unique (device, state, canonical attack) session is assigned entirely to train, validation, or test. This is stricter than blocked splitting because it prevents different windows from the same continuous capture session from appearing on both sides of the split. The resulting session allocation contains 2/1/1 benign sessions across train/validation/test, 18/6/6 reconnaissance sessions, 18/5/5 DoS sessions, and one training / one test session for each of Backdoor and Cryptojacking.

The test set contains 701 windows. Metrics are reported using accuracy, macro precision, macro recall, macro F1, and macro AUC. This protocol is the basis of the headline result because it is the most defensible estimate of generalization available from the current dataset.

B. Headline Session-Level Results

Table II reports the main result of the paper. Under the strict session-level holdout, the token-fusion transformer achieves 90.30% accuracy and 0.6952 macro F1. The macro scores are notably lower than the raw accuracy because the test set is dominated by Backdoor windows and because the Benign class fails completely under the cross-device condition discussed next.

The per-class breakdown in Table III explains where the aggregate metrics come from. Reconnaissance, Backdoor, and Cryptojacking are detected strongly. DoS achieves very high recall but weaker precision. The Benign class, however, is completely missed.

The benign failure is not random noise. All 66 benign test windows are misclassified, mostly as DoS. The reason

TABLE IV
EARLIER BLOCKED-SPLIT BINNED BENCHMARKS (SECONDARY CONTEXT)

Model	Accuracy	Macro F1
Majority baseline	43.17%	0.1206
RF	99.88%	0.9981
SVM	99.68%	0.9954
CNN-LSTM	99.08%	0.9802
Transformer fused	97.69%	0.9510

is a cross-device distribution shift: the benign training sessions come from EVSE-A, which has no host-event modality, while the held-out benign test session comes from EVSE-B, which does include host activity. The model therefore learns a misleading shortcut in which “benign” is associated with the absence of host evidence. This is an important paper result because it shows that the stricter split reveals a genuine deployment risk that easier splits hide.

C. Secondary Comparison with Earlier Blocked Results

Before the final session-level rewrite, we also evaluated earlier blocked-split baselines on the binned fused dataset. Those results are useful as secondary context because they show how easy it is to obtain inflated performance when train and test windows come from closely related capture structure. Table IV summarizes the earlier numbers.

These results are not wrong in the narrow sense of their own protocol, but they should not be read as the final paper headline. During the phase-3 audit, repeated near-perfect results were traced back to within-session similarity and session overlap. The session-level experiment therefore became the paper’s authoritative benchmark. The main lesson is methodological: on this dataset, a strong number is not enough unless the split makes memorization difficult.

D. Fusion and Explainability Insights

The earlier ablation results also showed that fusion is useful. Under the blocked split, the transformer dropped from 97.69% to 84.30% accuracy when reduced to a network-only variant, indicating that host evidence adds meaningful context even though it does not solve the harder session-level generalization problem on its own. Host-only and cross-attention variants collapsed, which we keep as negative results rather than hiding them.

The XAI artifacts complement the quantitative analysis. The SHAP feature map spans 18,944 attribution dimensions per prediction and highlights concrete host counters and network features rather than anonymous latent components. Representative host signals include `cache-misses`, `cpu-cycles`, context-switch indicators, block I/O events, and the `crypto_spec` counter that is directly relevant to cryptojacking behavior. Attention visualizations from the final encoder layer show that masked host regions are correctly suppressed when no host tokens are available, while active network regions receive concentrated weight around the most

informative temporal burst. These observations do not replace the class metrics, but they make the detector auditable and help explain why some attack classes remain easy while benign transfer does not.

E. Limitations

Two limitations shape how the results should be interpreted. First, the dataset itself provides only two Backdoor sessions and effectively one Cryptojacking capture source, so even a session-level evaluation remains bounded for these host-attack classes. Their high F1 scores should therefore be read as within-dataset evidence, not as proof of broad real-world generalization. Second, the benign cross-device failure shows that the model is sensitive to capture-environment shift. Future work should target domain adaptation, richer benign host coverage, and a cleaner benchmark in which all classes have multiple truly independent sessions.

VII. CONCLUSIONS

This paper presented the final phase-3 version of our EVSE anomaly detection study on CICEVSE2024. The completed system is an explainable transformer-based multi-modal detector that uses raw host and network event tokens instead of aggregated bins, fuses both modalities through self-attention, and produces both class predictions and explanation artifacts. Under the strict session-level holdout adopted for the final paper, the model achieved 90.30% accuracy and 0.6952 macro F1 on the full five-class task.

The most important outcome of the project is not only the final metric. It is the evaluation lesson that emerged during development. Easier blocked or random splits can make EVSE intrusion detection appear nearly solved, while a stricter session-level holdout reveals the remaining difficulties immediately. In our case, the benign cross-device failure became visible only after the split was tightened. At the same time, the model still showed that transformer-based fusion and explainability are viable and informative directions for EVSE security. The next step is therefore not to chase a cleaner number on the same split, but to build a more robust benchmark and a detector that can generalize across devices, states, and host-side capture conditions.

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